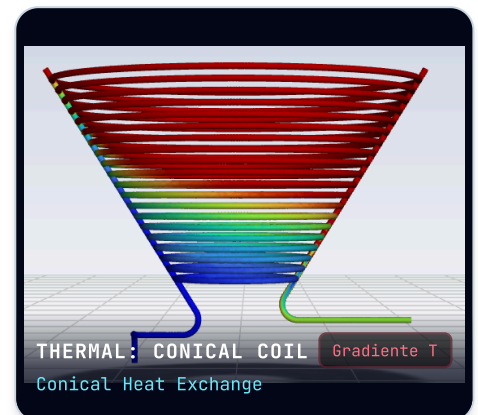
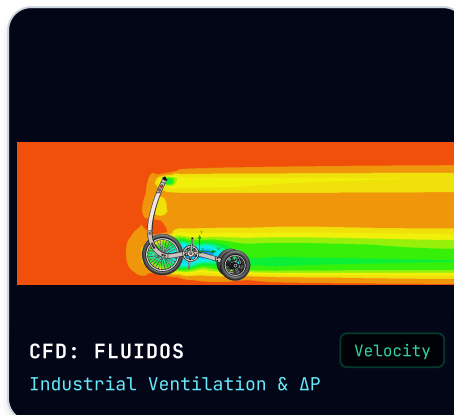
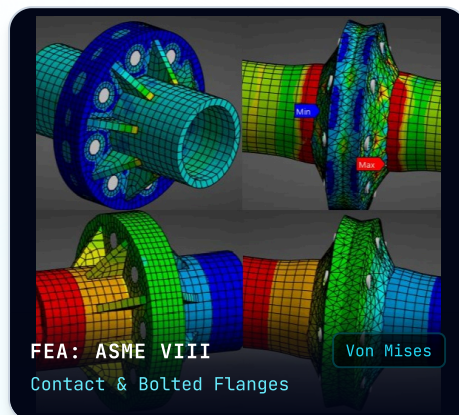


 Computational Numerical Simulation Consulting

Detail Design Backed by Numerical Simulation and Engineering Codes.

We design, evaluate and optimize mechanical and industrial systems through **CFD** and **FEA** in strict compliance with international standards before physical manufacturing or prototyping.



 CROSS-CUTTING CODES & STANDARDS

Certified Design & Technical Opinion

ASME BPVC
Sec. VIII Div. 1 & 2

ASME B31
B31.1 & B31.3 Piping

AISC 360
Structural Steel

AWS D1.1
Structural Welding

 SPECIALIZED SOFTWARE & CAE

High-Precision Simulation

ANSYS
Mechanical & Fluent

SOLIDWORKS
CAD 3D & FEA

INVENTOR
Autodesk Parametric

MATLAB
Numerical Computing

 CFD

Fluid flow, pressure drop & ventilation.

 FEA

Von Mises stress, fatigue & welds.

 THERMAL

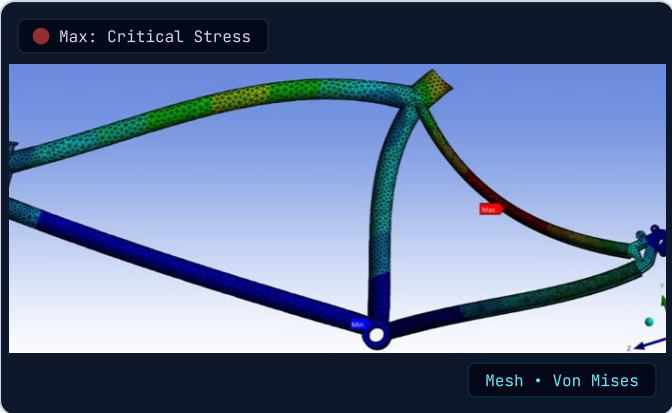
Heat exchangers & cooling.

 CAD 3D

Parametric 3D CAD & drafting.

Structural Analysis (FEM / FEA)

We validate **mechanical strength**, **stress concentrations**, **structural deflections**, and **weight reduction** of mission-critical components operating under severe static, cyclic, or dynamic conditions.



EXECUTED PROJECT: STRUCTURAL OPTIMIZATION

Von Mises Stress Analysis & Mass Reduction

Detailed mapping of peak (Max) and minimum (Min) stress zones across structural tubes and weld nodes. Mass optimization while safeguarding code safety factors and fatigue life.

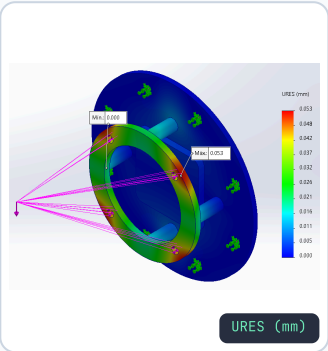
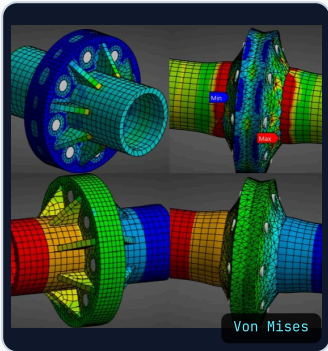
- ✓ Weight reduction
- ✓ Fatigue life assurance

EXECUTED PROJECT: VESSELS & FLANGED JOINTS

Non-Linear Contact, Gussets & Bolt Preload

Quasi-static simulation of bolted flanges with bolt pre-tension, elastomeric gaskets, and resulting displacement (URES) calculation compliant with ASME Sec. VIII.

- ✓ Sealing & gasket tightness
- ✓ Stiffness & URES Deflection



Linear Static
Stiffness & deflection



Non-Linear
Plasticidad y contacto



Dynamic & Modal
Vibration & seismic



Fatigue Life
Duty cycles & damage

ACCEPTANCE CRITERIA & ENGINEERING REPORTS

FEA Calculation Dossier

Safety Factors (SF)

Evaluation against yield strength (Sy) and ultimate tensile strength (Su).

Stress Linearization

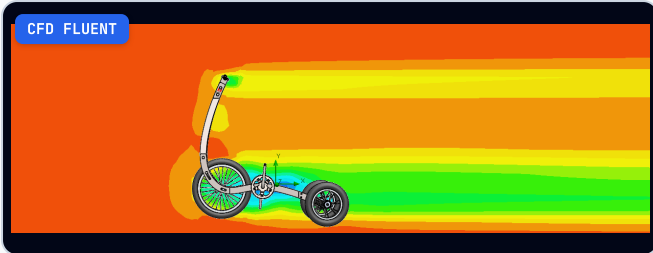
Decomposition into membrane, bending, and peak stresses per ASME standards.

Buckling & Stability

Bifurcation load multipliers and post-buckling non-linear stability analysis.

CFD Analysis & Heat Transfer

CFD FLUENT



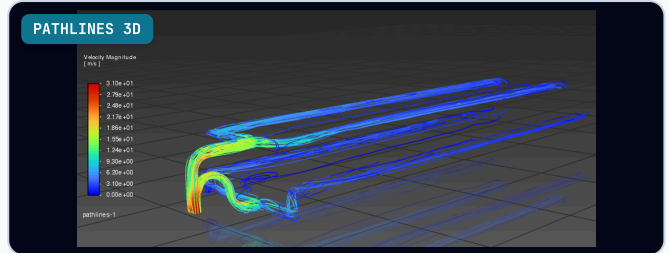
FLUID DYNAMICS

Ventilation & Air Quality

Air exchange simulation in industrial plants, contaminant dispersion, and thermal comfort evaluation.

Calidad de aire y ventilación industrial

PATHLINES 3D



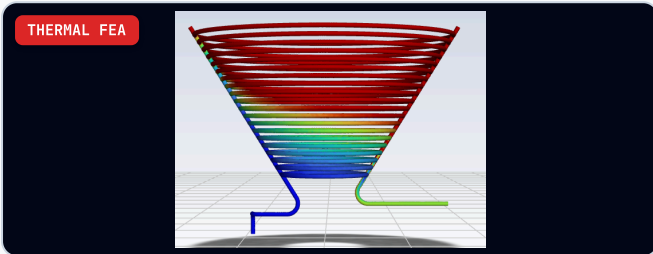
FLOW PATHLINES

Ductwork & Air Distribution

Geometry optimization to minimize pressure loss (ΔP), eliminate recirculation eddies, and balance branch lines.

Líneas de corriente y distribución en ductos

THERMAL FEA



HEAT TRANSFER

Conical Heat Exchange

Temperature gradients, thermal exchange efficiency, and global convective coefficients across fluid and wall.

Gradientes en intercambiadores de calor

MESH SURFACE



NUMERICAL PRE-PROCESSING

Surface Meshing & Convergence

Structured and unstructured high-fidelity meshes with boundary layers, ensuring stability and accuracy.

Choque y fatiga termo-mecánica

DMD ENGINEERING METHODOLOGY

Simulate • Analyze • Solve

01. PRE-PROCESS

Clean CAD geometry & high-fidelity mesh

02. PHYSICS

Real-world boundary conditions

03. SOLUTION

Navier-Stokes / FEM solvers

04. REPORT

Certified engineering sign-off & recommendations

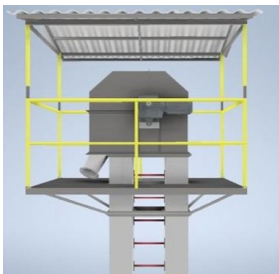
Descriptive calculation memory

3D flow animations & contours

Native numerical simulation files

Standardized technical drawings

3D CAD Modeling & Industrial Scope

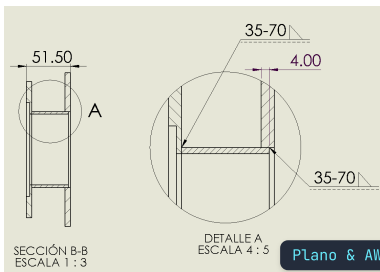


PARAMETRIC 3D MODELING

Machine Design, Access Platforms & Reverse Engineering

We develop complex mechanical assemblies, OSHA-compliant maintenance platforms, and dimensionally accurate 3D reverse engineering ready for simulation.

- ✓ Parametric 3D solid models
- ✓ Specialized engineering support



MANUFACTURING DOCUMENTATION

Workshop Drawings, Part Breakdown & AWS Welding

We generate detailed fabrication drawings with section views, ISO dimensional tolerances, and complete AWS D1.1 structural welding symbology.

- ✓ Standardized DWG / PDF drafting
- ✓ AWS welding symbology

KEY INDUSTRIAL SECTORS & APPLICATIONS

Tailored Solutions



Process Equipment

Reactors, boilers, and heat exchangers, industrial furnaces, and pressure vessels under ASME codes.



Oil & Gas / Piping

ASME B31 piping networks, leak-tight flanged assemblies, manifolds, and control valves.



Mining & Structures

Transfer chutes, hoppers, heavy-duty mobile chassis, and welded frames under AWS standards.



HVAC & Naves

Industrial ventilation, dust extraction, contaminant dispersion, and indoor thermal comfort.

PRECISION CONSULTING ENGINEERING

WE DELIVER TAILORED SOLUTIONS FOR YOUR NEEDS • CONTACT US!

✉ comercial@dmd-engnsolutions.com

🌐 www.dmd-engnsolutions.com

📍 Domestic & International Services